

HAR versus Deep and Graph-Attention Models for Multi-Horizon Volatility Forecasting: A Panel-Correct Study on Vietnamese and U.S. Equities

Author One¹ and Author Two¹

Affiliation, City, Country
`{author1,author2}@example.edu`

Abstract. We study whether deep sequence models or a cross-sectional graph-attention branch can improve daily Parkinson-variance forecasting over the Heterogeneous Autoregressive (HAR) baseline, and we evaluate them with panel-correct inference. Three equity panels are used: VN30 (33 tickers) and VN100 (104 tickers) as the Vietnamese case study, and a long-history S&P 500 subset (457 tickers) as a data-rich cross-market check. Against HAR and GARCH(1,1) we benchmark a pooled LSTM over the three HAR features, an LSTM with a Graph Attention Network branch over a graphical-lasso partial-correlation graph, a frozen-expert convex HAR-deep combination, and HAR-anchored residual models, at horizons $\{1, 5, 10, 22\}$ with five seeds. Significance is assessed with the date-clustered Diebold–Mariano test (one loss differential per calendar date); a naive per-observation test on a cross-sectionally dependent panel overstates significance by a factor of order $\sqrt{N}$. Three findings follow. First, HAR is very hard to beat on the small Vietnamese panels: under date-clustered inference no model, including the forecast combination, significantly beats HAR at any horizon (closest $p = 0.08$). Second, on the large S&P 500 the deep temporal LSTM (no graph) significantly beats HAR, by up to +7.1% QLIKE at $h22$ ($p < 0.001$), and the convex combination beats HAR at all four horizons ($p < 0.001$); competitiveness grows with panel size. Third, the cross-sectional graph adds no out-of-sample value on any panel: removing it never improves on the no-graph model, a leakage-safe linear neighbour-signal regressor adds essentially zero incremental out-of-sample R^2 , and diagnostics rule out a bug or overfitting. We report MSE, RMSE, MAE, QLIKE and R^2 together, since the ranking is metric- and panel-dependent.

Keywords: Volatility forecasting · HAR · LSTM · Graph Attention Network · Diebold–Mariano.

1 Introduction

Daily volatility forecasting underpins risk management, position sizing and derivative pricing. The HAR model of Corsi [1] is a strong, parsimonious benchmark:

three OLS coefficients on daily, weekly and monthly aggregates capture much of the long-memory structure. Deep sequence models and graph neural networks are frequently proposed to capture nonlinear temporal dynamics and cross-sectional spillovers, but the evidence that they beat HAR out of sample is mixed and positive claims are often confounded by data-design choices.

We ask two questions, with the Vietnamese equity market as the primary case study and a long-history U.S. S&P 500 subset as a data-rich cross-market check: (1) Can deep or HAR-anchored models — a pooled LSTM over the three HAR features, a convex HAR-deep combination, or HAR-residual correctors — beat HAR and GARCH out of sample under panel-correct inference, and at which horizons? (2) Does adding a cross-sectional Graph Attention Network branch, with edges estimated by graphical lasso, contribute any out-of-sample value beyond the same-capacity no-graph model? Our contributions are: (i) a panel-correct multi-horizon benchmark of HAR against an LSTM, an LSTM-GAT, a convex combination and HAR-anchored residual models on three panels, with significance by the date-clustered Diebold–Mariano test (a naive per-observation test overstates significance by a factor of order $\sqrt{N}$ on this cross-sectionally dependent panel); (ii) the main finding that HAR is not significantly beaten on the small Vietnamese panels, whereas on the data-rich S&P 500 the deep temporal LSTM significantly beats HAR (up to +7.1% QLIKE at $h22$) and the convex combination beats it at every horizon, so deep-versus-HAR competitiveness scales with data; (iii) a model-free demonstration that the cross-sectional graph adds no out-of-sample value on any panel, a genuine null rather than a bug or overfitting; and (iv) per-metric fairness — reporting MSE, RMSE, MAE, QLIKE and R^2 together and showing the ranking is metric- and panel-dependent.

2 Related Work

HAR and classical models. HAR [1] approximates the long memory of realized volatility with a daily/weekly/monthly cascade; recent work reports that rolling-window and specification choices, rather than model class, often drive apparent gains over HAR [12]. HARQ [9] exploits measurement error, and GARCH-family models remain standard conditional-variance benchmarks. We use HAR and GARCH(1,1) as the baselines to beat. **Deep and graph models.** LSTMs [3] are widely applied to financial series, and graph neural networks model cross-asset spillovers, e.g. Graph Attention Networks [4], multivariate realized-volatility GNNs [10] and spillover-aware GNN-HAR hybrids [11]; evidence is mixed. **Graph construction.** Rather than a market-factor-dominated correlation graph, we estimate a sparse partial-correlation graph with graphical lasso [5] on training rows only. **Evaluation.** Volatility is latent, so we use the proxy-robust QLIKE loss [8] alongside MSE, with significance by the Diebold–Mariano test [6] and the Harvey–Leybourne–Newbold small-sample correction [7].

3 Method

Target and features. The target is the Parkinson variance [2] at day $t+h$ for horizons $h \in \{1, 5, 10, 22\}$ (the primary date-clustered study; the descriptive per-observation and snapshot studies use $h \in \{1, 5\}$), a non-negative point forecast. Following HAR, three features per node are used: the daily Parkinson variance at t , its 5-day rolling mean (weekly) and its 22-day rolling mean (monthly). The same features drive every model, so differences reflect functional form, not the information set.

Main model: per-observation LSTM. The main model (“LSTM”) is a two-layer LSTM (hidden 64) over the lookback window of the three features; a single pooled model is trained over all tickers, each standardized by its own scaler fit on training rows only, with a linear output inverse-transformed for evaluation. Overfitting is controlled by dropout, weight decay, gradient clipping, a ReduceLROnPlateau schedule and early stopping. Training examples are formed by *per-observation pooling*: every (ticker, window) pair is an independent example and the chronological 80/10/10 split is applied within each ticker’s own history, yielding tens of thousands of windows and interleaving each ticker’s regimes.

Graph variant: LSTM+GAT (ablation). To test whether a cross-sectional graph helps, a Graph Attention Network branch [4] is added (Fig. 1): a per-node LSTM encodes the temporal window, and a GAT reads the node features at t and attends over a fixed Top-5 graphical-lasso partial-correlation graph estimated on training rows only; the branches are concatenated into an MLP head. The leave-one-out “LSTM (w/o GAT)” removes the spatial branch. A GAT operates over a common-date cross-section, so LSTM+GAT requires a *common-date snapshot* design (only dates on which all nodes are present) with a single global 80/10/10 split; we thus report the graph ablation as a separate study to avoid confounding.

Baselines. HAR is a pooled OLS on the three features refit on the same training rows; GARCH(1,1) is a per-ticker conditional-variance model.

Protocol. Each configuration runs five seeds {42, 123, 2026, 7, 2024}, up to 20 epochs with early stopping. We report MSE, RMSE, MAE, QLIKE and R^2 , seed-averaged, with a shared QLIKE floor 10^{-8} ; the primary decision metric is QLIKE. Predictive-accuracy tests are run only on the proxy-robust losses — QLIKE (primary) and squared error; MAE is reported as a descriptive metric but not tested, because absolute error is not robust to noise in the volatility proxy and can invert forecast rankings [8]. Significance is assessed with the *date-clustered* Diebold–Mariano test: the per-observation loss differential is collapsed to one value per calendar date before the statistic (HLN correction, HAC lag $h-1$) is computed. This is the panel-correct treatment for a cross-section that shares each trading date; a naive per-observation test treats the effective sample as the number of tickers times the number of dates and inflates the statistic by a factor of order $\sqrt{N}$ (roughly six on VN30, ten on VN100). All significance statements use the date-clustered test unless a result is explicitly labelled per-observation. In the results tables a compact marker (*) flags a model that significantly beats HAR on QLIKE under the date-clustered test ($p < 0.05$); the per-horizon DM

p -values versus HAR are collected in the HAR-anchored tables (Tables 3, 4), the single headline comparison against the benchmark.

4 Data

Three panels are used with the Parkinson target and three HAR features: **VN30** (33 tickers, primary, Vietnam), **VN100** (104, vnstock), and **S&P 500** (~ 500 , U.S. robustness; Yahoo-sourced, only aggregate metrics stored). The main study uses per-ticker 80/10/10 splits and a pooled model; the graph study uses common-date fixed-node snapshots with a global split. The S&P 500 list is current membership, so U.S. levels carry survivorship bias; we claim the short-versus-long ordering transfers, not absolute levels.

5 Experiments

We run five studies on the same features across VN30, VN100 and the S&P 500 subset: (1) the *primary* HAR-anchored ladder E0–E10 under date-clustered inference at $h \in \{1, 5, 10, 22\}$ (Sec. 6.4), the source of the headline results; (2) a *design comparison*, LSTM vs HAR and GARCH on the per-observation design (lookback 10; $h \in \{1, 5\}$), reporting point estimates and naive per-observation statistics; (3) a *graph-check* ablation, LSTM+GAT vs LSTM (w/o GAT) on the snapshot design; (4) *lookback* 10 vs 22 (snapshot, VN30); and (5) a *model-free graph screen* of whether any cross-stock neighbour signal adds out-of-sample value beyond HAR.

6 Results

Parkinson variance is of order 10^{-7} ; MSE, RMSE, MAE are scaled ($\text{MSE} \times 10^{-7}$, $\text{RMSE/MAE} \times 10^{-4}$). Row order is HAR $\rightarrow$ GARCH $\rightarrow$ LSTM (and, for the graph study, adding LSTM (w/o GAT) $\rightarrow$ LSTM+GAT). All values are five-seed test means.

6.1 Main study — per-observation LSTM (descriptive point estimates)

Table 1 reports QLIKE point estimates for this descriptive design comparison. On the Vietnamese panels the LSTM’s QLIKE point estimates edge below HAR at the short horizons; a naive per-observation DM on this design flags several as significant, but that test overstates significance on a cross-sectionally dependent panel, and under the panel-correct date-clustered test (Sec. 6.4) none of these Vietnamese gaps is significant. Every learned model beats GARCH by large margins. On the S&P 500 the LSTM attains both the lower QLIKE and lower MSE at both horizons, and the date-clustered study (Sec. 6.4) confirms the S&P 500 beat-HAR result is statistically significant there. The panel-correct verdicts are in Sec. 6.4.

Table 1. Main study (per-observation, per-stock 80/10/10): QLIKE point estimates (lower is better; 5-seed mean; best in bold). This is a descriptive design comparison; panel-correct significance is not marked here, because the date-clustered DM verdicts are computed on the snapshot design (Table 3, Sec. 6.4). A naive per-observation DM on this design flags the short-horizon Vietnamese LSTM-vs-HAR gaps as significant, but overstates significance on a cross-sectionally dependent panel by a factor of order $\sqrt{N}$ (Method).

Panel	h	HAR	GARCH	LSTM
VN30	1	0.4675	0.7434	0.4578
VN30	5	0.5514	0.7409	0.5458
VN100	1	0.4798	0.7031	0.4735
VN100	5	0.5441	0.6993	0.5372
S&P500	1	0.3616	0.7286	0.3654
S&P500	5	0.4226	0.7287	0.4232

Table 2. Graph-check (common-date snapshot, lookback 10): QLIKE (5-seed mean; best in bold) and the leave-one-out Diebold–Mariano statistic (DM_{LOO}) comparing LSTM+GAT with LSTM(w/o GAT) on QLIKE (*: $p < 0.05$; a positive value favours the no-graph model). The graph never improves QLIKE.

Panel	h	HAR	GARCH	w/o GAT	LSTM+GAT	DM_{LOO}
VN30	1	0.3946	0.6500	0.4120	0.4528	+6.40*
VN30	5	0.4531	0.6420	0.4663	0.4991	+4.94*
VN100	1	0.4843	0.6210	0.5204	0.5296	+0.93
VN100	5	0.5442	0.6157	0.5552	0.5588	+0.38
S&P500	1	0.3390	0.3842	0.3401	0.3473	+3.23*
S&P500	5	0.3680	0.3890	0.3582	0.3701	+6.87*

6.2 Graph-check ablation

Table 2 gives the snapshot QLIKE and the leave-one-out DM. Removing the GAT branch improves QLIKE in all eight configurations (LSTM (w/o GAT) vs LSTM+GAT favors the no-graph model everywhere, significant on VN30 and S&P 500), so the graph does not help; at best it is QLIKE-neutral on VN100. Under this design HAR is strongest on the Vietnamese panels, while on the large S&P 500 the price-only LSTM (w/o GAT) attains the best QLIKE at $h5$ (0.3582 vs HAR 0.3680).

6.3 Lookback variation and data scaling

On VN30 the longer lookback (22 vs 10) does not help: LSTM (w/o GAT) QLIKE is essentially unchanged (0.4137 vs 0.4120 at $h1$) and LSTM+GAT is slightly worse (0.4714 vs 0.4528); the graph again hurts (DM +8.80 at $h1$). Reading QLIKE across panels of increasing size, under the panel-correct date-clustered inference of Sec. 6.4, shows a consistent data-scaling gradient: HAR is

not significantly beaten on VN30 or VN100 at any horizon, whereas on the large S&P 500 subset the deep temporal LSTM significantly beats HAR at every horizon (dQLIKE +3.01% at $h1$ to +7.10% at $h22$, $p < 0.001$ except $h10$ at $p = 0.001$) and the convex combination beats HAR at every horizon ($p < 0.001$). The significant beat-HAR result thus emerges only once the panel is large and data-rich, and it holds under the same inference convention under which the Vietnamese panels show no significant win.

6.4 HAR-anchored residual and forecast-combination study (E0–E10)

A complementary question to “does a full-target deep model beat HAR” is whether HAR-*anchored* learning — combining or correcting HAR with a deep expert rather than replacing it — improves on HAR, and whether the cross-sectional graph adds anything once HAR carries the level. We evaluate an eleven-rung ladder: **E0** HAR; **E1/E2** full-target LSTM / LSTM+GAT; **E3** a frozen-expert convex combination of HAR and the deep expert with a single mixing weight fit on the validation predictions only; **E5/E6/E7** additive HAR-residual models whose deep branch (LSTM-only, the graph/GAT branch, or the combined branch) predicts the HAR residual with a zero-initialised head (equal to HAR at initialisation); **E8** a multiplicative HAR-anchored residual $(\text{HAR} + \varepsilon) \cdot \exp(\text{corr})$, positive and bounded by construction; and **E9/E10** static and dynamic (regime-aware) gates scaling the residual correction. The design is the common-date snapshot design of the graph study, extended to $h \in \{1, 5, 10, 22\}$, with a target-overlap purge of h snapshots at each boundary, a per-horizon pooled-OLS HAR anchor fit on training rows only, and residual targets built from expanding-window *cross-fitted* HAR predictions (out-of-sample residuals, not in-sample). Five seeds; primary metric QLIKE. Significance uses the **date-clustered** Diebold–Mariano test (one loss differential per calendar date), the panel-correct treatment for a cross-section that shares each date.

Table 3 summarises QLIKE for HAR and the three leading learned rungs on the Vietnamese panels; the full E0–E10 ladder, with out-of-sample R^2 relative to HAR and the date-clustered DM p -value versus HAR, is in Table 5 (Appendix). **No model significantly beats HAR on VN30 or VN100 at any horizon:** the convex combination (E3), the closest competitor, has date-clustered $p \in 0.14\text{--}0.51$ (VN30) and $0.08\text{--}0.29$ (VN100), never below 0.05; the residual and gated variants have larger p -values still. Point-estimate QLIKE gaps favour the hybrids at longer horizons (e.g. E3 reaches +5.85% QLIKE at VN100 $h22$, 0.5816 vs HAR 0.6177), but they are within noise given the short common-date test windows (of order 130 dates on VN30, 50 on VN100). Full-target deep models (E1/E2) underperform HAR at short horizons ($h1$: VN30 $p < 0.001$; VN100 $p = 0.023/0.035$), whereas the HAR-anchored variants track HAR closely at short horizons and never trail far, i.e. anchoring removes the short-horizon penalty.

Large-panel result (S&P 500). On the data-rich S&P 500 subset (457 nodes, ~ 300 test dates; Table 4) the same evaluation gives the opposite verdict for the deep temporal model: the full-target LSTM (E1) significantly beats HAR

Table 3. HAR-anchored ladder on the Vietnamese panels (common-date snapshot; 5-seed means): QLIKE for HAR (E0), the full-target deep models (E1 LSTM, E2 LSTM+GAT) and the convex combination (E3, the closest competitor to HAR), with the date-clustered DM p of E3 vs HAR (lower QLIKE is better). No rung beats HAR at any horizon, so no significance marker (*, date-clustered $p < 0.05$ vs HAR) appears; the full detail (E0–E10 with R^2_{OOS} and per-rung DM p) is in Table 5 (Appendix).

Panel	h	HAR	E1 LSTM	E2 LSTM+GAT	E3 convex	DM p (E3 vs HAR)
VN30	1	0.3946	0.4383	0.4915	0.3984	0.237
VN30	5	0.4531	0.4810	0.5185	0.4639	0.135
VN30	10	0.4849	0.5291	0.5394	0.4958	0.322
VN30	22	0.5025	0.5185	0.5603	0.5137	0.511
VN100	1	0.4844	0.5371	0.5207	0.4821	0.242
VN100	5	0.5441	0.5614	0.5582	0.5339	0.294
VN100	10	0.5985	0.5750	0.5803	0.5701	0.202
VN100	22	0.6177	0.5888	0.6031	0.5816	0.078

at every horizon (dQLIKE +3.01%/+4.09%/+4.18%/+7.10% at $h1/h5/h10/h22$; $p < 0.001$ except $h10$ at $p = 0.001$), and the convex combination (E3) beats HAR at every horizon (+3.16% to +5.39%, $p < 0.001$). The graph-carrying full model (E2) is weaker than E1 at every horizon and fails to beat HAR at $h1$ and $h10$. This beat-HAR result holds under the same date-clustered inference under which the Vietnamese panels show no significant win, so the difference is one of data scale, not of inference convention; on this panel the LSTM also leads on RMSE and MAE, not only QLIKE. The additive/graph residual rungs (E5–E8) are numerically fragile on this large cross-section (e.g. E5 QLIKE 2.9978 at $h22$, E7 QLIKE 5.7090 at $h10$, vs HAR 0.4596/0.4303), so the clean beat-HAR models on the large panel are E1 and E3.

Graph attribution. A graph contribution requires the graph/GAT residual (E6) or the combined residual (E7) to beat the same-capacity no-graph residual (E5), not merely HAR. Under the paired date-clustered DM, in no panel and at no horizon does the graph residual significantly beat the no-graph residual: on the Vietnamese panels none of the eight paired contrasts approaches significance (all $p \geq 0.41$, sign mixed), while on the S&P 500 the no-graph residual is favoured at every horizon, significantly at $h1$ ($p = 0.0003$) and $h10$ ($p = 0.015$). The full per-horizon paired p -values are in the results record (`reports/experiment_results.md`). This is consistent with the leave-one-out ablation.

Model-free graph screening. Because the ablations compare specific architectures, we additionally rule out cross-stock signal model-free. Appending a neighbour-derived signal to the three HAR features in a train-only OLS and measuring the incremental test R^2 over HAR upper-bounds what any model consuming those edges could extract linearly (leakage-safe; edges train-only). Across VN30, VN100 and the S&P 500 subset at every horizon the incremental test R^2 is at most about 1% and near zero for the theoretically decisive

Table 4. HAR-anchored ladder on the S&P 500 subset (457 nodes; 5-seed means, date-clustered DM p vs HAR). * marks a model that significantly beats HAR on QLIKE ($p < 0.05$). Non-fragile rungs shown; the additive/graph residual rungs E5–E8 are numerically fragile on this cross-section and are discussed in the text.

h	Model	QLIKE	dQLIKE% vs HAR	DM p (date-clust.)
1	E0 HAR	0.3776	—	—
1	GARCH	0.4843	−28.27	< 0.001
1	E1 LSTM (full)	0.3662*	+3.01	< 0.001
1	E2 LSTM+GAT (full)	0.3826	−1.32	0.305
1	E3 convex combination	0.3656*	+3.16	< 0.001
5	E0 HAR	0.4158	—	—
5	GARCH	0.4830	−16.16	< 0.001
5	E1 LSTM (full)	0.3988*	+4.09	< 0.001
5	E2 LSTM+GAT (full)	0.4078*	+1.93	0.005
5	E3 convex combination	0.4009*	+3.58	< 0.001
10	E0 HAR	0.4303	—	—
10	GARCH	0.4842	−12.53	< 0.001
10	E1 LSTM (full)	0.4123*	+4.18	0.001
10	E2 LSTM+GAT (full)	0.4230	+1.69	0.115
10	E3 convex combination	0.4147*	+3.62	< 0.001
22	E0 HAR	0.4596	—	—
22	GARCH	0.4883	−6.23	0.121
22	E1 LSTM (full)	0.4270*	+7.10	< 0.001
22	E2 LSTM+GAT (full)	0.4386*	+4.59	0.004
22	E3 convex combination	0.4349*	+5.39	< 0.001

innovation and directed lead-lag screens; the only small positives are contemporaneous neighbour-level signals that sit close to a shuffled-edge placebo. A single leakage-safe regression of the target on HAR plus the mean Parkinson variance of a node’s Top-5 neighbours at day t — bypassing the GAT entirely — yields incremental out-of-sample R^2 of essentially zero everywhere (VN30 −0.0001/+0.0013/−0.0014/−0.0004; VN100 −0.0007/+0.0091/−0.0010/+0.0007). Diagnostics confirm a genuine null: the graphical-lasso edge set barely persists train→test (Top-5 Jaccard 0.09–0.17), the GAT branch is alive (a ~15%-of-HAR correction at VN100 $h22$) yet does not beat HAR, its attention collapses to a near-uniform neighbour average (entropy ≈ 0.996), and the graph residual shows no train-vs-test overfitting gap.

Relation to the per-observation tables. The per-observation Diebold–Mariano statistics in the main study (Table 1) flag short-horizon LSTM-vs-HAR gaps on the Vietnamese panels as significant, but that test overstates significance on this cross-sectionally dependent panel (Sec. 6.4); under the date-clustered inference used throughout, those Vietnamese gaps are not significant. The point estimates are unchanged, symmetrically for the cells where HAR nominally beats the deep model; the one beat-HAR result that survives date-clustered inference is on the large S&P 500 subset (Table 4), not on the Vietnamese panels.

7 Discussion

HAR is hard to beat on small panels; data scale unlocks a deep advantage. Under panel-correct date-clustered inference no model significantly beats HAR on VN30 or VN100 at any horizon, while the same architectures, evaluated identically, significantly beat HAR at every horizon on the large S&P 500 subset (the LSTM improves QLIKE by up to +7.1% at $h22$; the convex combination beats HAR at every horizon). Because the inference convention is held fixed across panels, the difference is one of data scale: a flexible model needs a large, data-rich cross-section to out-of-sample beat a parsimonious baseline, and HAR’s edge on the small Vietnamese panels is a small-sample phenomenon rather than a fundamental ceiling. **The graph adds no out-of-sample value, and this is a genuine null.** No graph-carrying model beats its same-capacity no-graph counterpart on any panel (leave-one-out ablation, paired residual attribution, and the S&P 500 case where the no-graph residual is significantly favoured); the model-free screen shows a leakage-safe linear neighbour-signal regressor adds ≈ 0 incremental out-of-sample R^2 , and diagnostics rule out a bug or overfitting (edge set does not persist train $\rightarrow$ test, attention collapses to a uniform average, no train-vs-test overfitting gap). The cross-sectional dependence the graph encodes is dominated by a common market factor already captured by each ticker’s own HAR persistence. **Data design and inference convention both shape the apparent verdict.** Two secondary observations support this reading: the deep-vs-HAR point estimates depend on the data design (the same LSTM’s estimates favour it under the richer per-observation design but not under the common-date snapshot), and, more importantly, the significance verdict depends on the inference convention (a naive per-observation test inflates the statistic by $\sim \sqrt{N}$). Both caution against reading a beat-HAR claim from a favourable design plus a naive test. **Metrics.** We report all five metrics rather than the one that flatters a model; on the S&P 500 the deep temporal model leads on QLIKE, RMSE and MAE simultaneously, while on the Vietnamese panels the best metric differs by model (HAR on QLIKE and squared error, the multiplicative HAR-anchored residual on MAE).

8 Limitations

The graph-defined snapshot design retains on the order of 130 test dates on VN30 and 50 on VN100; under date-clustered inference these are the effective sample sizes, so even sizeable long-horizon point-estimate gaps (e.g. the convex combination’s +5.85% at VN100 $h22$) are not significant — a “not demonstrated” rather than “no effect at any sample size” result. Model selection uses validation QLIKE for the HAR-anchored study and validation MSE for the main-study point-estimate tables; we report all five metrics rather than cherry-pick. The graph is only evaluable on the data-poorer snapshot design, though the negative result rests on within-design comparisons plus the design-independent model-free screen; the graph consumed here is the binary support of the graphical-lasso

graph, and while the screen additionally tests signed, weighted, innovation and directed lead-lag signals, volume-shock and sector graphs remain follow-up work. GAT attention is quadratic in nodes (small batch needed at ~ 500 nodes). The S&P 500 panel is the long-history subset of current constituents (survivorship bias); we claim the data-scaling direction and the beat-HAR-under-scale result on this subset, not the full index or the absolute level. Vietnam is the case study and the S&P 500 subset the cross-market check; no universal generality is claimed, and a masked-panel robustness check that relaxes the common-date intersection is ongoing and not included here. All numbers are read from stored results.

9 Conclusion

Evaluated with panel-correct date-clustered Diebold–Mariano inference on three equity panels, HAR is very hard to beat on the small Vietnamese markets: no model — LSTM, LSTM+GAT, convex combination or HAR-anchored residual — significantly beats HAR on VN30 or VN100 at any horizon. On the large, data-rich S&P 500 subset the same evaluation shows the deep temporal LSTM (no graph) significantly beating HAR at every horizon, by up to +7.1% QLIKE at $h22$ ($p < 0.001$), with the convex HAR–deep combination beating HAR at every horizon; every learned model beats GARCH. Deep-versus-HAR competitiveness therefore grows with panel size. Adding a Graph Attention Network over a graphical-lasso graph contributes no out-of-sample value on any panel — a genuine null established by leave-one-out and paired-residual ablations and by a model-free neighbour-signal screen, with diagnostics ruling out a bug or overfitting. We recommend a graph-free HAR or HAR-anchored deep model: HAR alone on small panels, where it is not significantly beaten, and a deep temporal or convex-combination model on large, data-rich panels, where it significantly improves on HAR.

A Full HAR-anchored ladder tables (VN30, VN100)

Table 5 gives the complete eleven-rung ladder (E0–E10) on the Vietnamese panels, summarised in Table 3 of the main text.

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LSTM+GAT: LSTM temporal branch + GAT spatial branch (graphical-lasso graph) -> head; ablation removes the GAT branch (main LSTM).

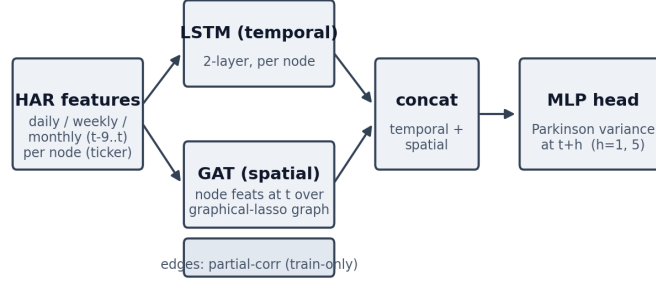


Fig. 1. LSTM+GAT: a per-node LSTM temporal branch and a GAT spatial branch over a graphical-lasso partial-correlation graph are concatenated into an MLP head; the leave-one-out ablation removes the GAT branch to give the main LSTM.

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Table 5. Full HAR-anchored ladder (E0–E10) on VN30 and VN100 (common-date snapshot; 5-seed means): QLIKE (lower better), out-of-sample R^2 vs HAR, and date-clustered DM p vs HAR. n/a: the static gate has no per-observation differential from HAR. Summarised in Table 3.

h	Model	VN30			VN100		
		QLIKE	R^2_{OOS}	DM p	QLIKE	R^2_{OOS}	DM p
1	E0 HAR	0.3946	0.0000	—	0.4844	0.0000	—
1	E1 LSTM (full)	0.4383	−0.1313	< 0.001	0.5371	−0.1056	0.023
1	E2 LSTM+GAT (full)	0.4915	−0.2438	< 0.001	0.5207	−0.0599	0.035
1	E3 convex combination	0.3984	−0.0017	0.237	0.4821	0.0027	0.242
1	E5 add. resid. (LSTM)	0.3946	−0.0001	0.875	0.4842	0.0001	0.782
1	E6 add. resid. (graph)	0.3946	−0.0001	0.882	0.4842	0.0002	0.603
1	E7 add. resid. (both)	0.3946	0.0000	0.795	0.4842	0.0002	0.611
1	E8 mult. HAR-anchored	0.3947	−0.0002	0.894	0.4879	−0.0058	0.582
1	E9 static gate	0.3946	0.0000	n/a	0.4844	0.0000	n/a
1	E10 dynamic gate	0.3946	0.0000	0.754	0.4843	0.0000	0.601
5	E0 HAR	0.4531	0.0000	—	0.5441	0.0000	—
5	E1 LSTM (full)	0.4810	−0.0803	0.004	0.5614	−0.0153	0.244
5	E2 LSTM+GAT (full)	0.5185	−0.1603	< 0.001	0.5582	0.0027	0.415
5	E3 convex combination	0.4639	−0.0171	0.135	0.5339	0.0208	0.294
5	E5 add. resid. (LSTM)	0.4531	−0.0001	0.994	0.5439	0.0002	0.746
5	E6 add. resid. (graph)	0.4531	−0.0001	0.926	0.5438	0.0003	0.634
5	E7 add. resid. (both)	0.4532	−0.0002	0.906	0.5438	0.0003	0.631
5	E8 mult. HAR-anchored	0.4531	−0.0000	0.954	0.5480	0.0072	0.823
5	E9 static gate	0.4531	0.0000	n/a	0.5441	0.0000	n/a
5	E10 dynamic gate	0.4531	−0.0000	0.932	0.5441	0.0001	0.624
10	E0 HAR	0.4849	0.0000	—	0.5985	0.0000	—
10	E1 LSTM (full)	0.5291	−0.1147	0.110	0.5750	0.0433	0.283
10	E2 LSTM+GAT (full)	0.5394	−0.1221	0.002	0.5803	0.0428	0.455
10	E3 convex combination	0.4958	−0.0244	0.322	0.5701	0.0441	0.202
10	E5 add. resid. (LSTM)	0.4853	−0.0002	0.782	0.5979	0.0003	0.673
10	E6 add. resid. (graph)	0.4854	−0.0003	0.560	0.5883	0.0119	0.458
10	E7 add. resid. (both)	0.4854	−0.0003	0.588	0.5929	0.0062	0.459
10	E8 mult. HAR-anchored	0.4850	−0.0001	0.851	0.6008	0.0160	0.921
10	E9 static gate	0.4849	0.0000	n/a	0.5884	0.0118	0.488
10	E10 dynamic gate	0.4850	−0.0001	0.600	0.5966	0.0020	0.440
22	E0 HAR	0.5025	0.0000	—	0.6177	0.0000	—
22	E1 LSTM (full)	0.5185	−0.0609	0.527	0.5888	0.0556	0.164
22	E2 LSTM+GAT (full)	0.5603	−0.1141	0.073	0.6031	0.0523	0.540
22	E3 convex combination	0.5137	−0.0416	0.511	0.5816	0.0512	0.078
22	E5 add. resid. (LSTM)	0.5026	−0.0005	0.913	0.6171	0.0005	0.765
22	E6 add. resid. (graph)	0.5026	−0.0005	0.895	0.5925	0.0374	0.481
22	E7 add. resid. (both)	0.5027	−0.0006	0.877	0.5922	0.0404	0.555
22	E8 mult. HAR-anchored	0.5027	−0.0010	0.816	0.6187	0.0065	0.973
22	E9 static gate	0.5025	0.0000	n/a	0.5912	0.0430	0.569
22	E10 dynamic gate	0.5025	−0.0001	0.896	0.6043	0.0180	0.480